

SolarGrid Demo Utility — Plan Change Impact Assessment Procedure

SG-MON-007

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

Document ID	SG-MON-007
Document Title	SolarGrid Demo Utility — Plan Change Impact Assessment Procedure
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define monitoring events, material-change thresholds, impact assessment, plan validity, and human review.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Scope: Define monitoring events, material-change thresholds, impact assessment, plan validity, and human review.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This procedure defines the synthetic monitoring and plan-change workflow used to determine whether an approved operating plan remains valid after a material system change.

Thresholds are configurable demo values. A deploying organization can replace them with approved operating and planning criteria without changing the workflow architecture.

2. Baseline Operating Plan

MON-001 — Each approved plan shall have a unique plan identity, creation timestamp, validity interval, operating snapshot reference, and approval status.

MON-002 — The baseline plan shall record the expected solar forecast, load, generator availability, BESS availability, network topology, reserve requirement, and key constraint margins.

3. Current System Snapshot

The current system snapshot contains the latest known values used to compare actual or newly reported conditions against the baseline plan.

MON-003 — A material plan-impact comparison shall use a current system snapshot rather than only the original planning inputs.

4. Material Change Thresholds

Change Type	Synthetic Trigger Threshold	Impact Assessment
Solar forecast	Absolute change >15%	Mandatory
Load	Absolute change >10%	Mandatory
Generator availability	Change of >1 unit	Mandatory
BESS availability	Capability change >10 MW	Mandatory
Network topology	Any topology change	Mandatory

MON-004 — A solar forecast change greater than 15% shall trigger plan-impact assessment.

MON-005 — A load change greater than 10% shall trigger plan-impact assessment.

MON-006 — A generator availability change involving more than one unit shall trigger plan-impact assessment.

MON-007 — A BESS availability change greater than 10 MW shall trigger plan-impact assessment.

MON-008 — Any network topology change shall trigger mandatory plan-impact assessment.

5. Change Impact Assessment

MON-009 — The impact assessment shall identify affected constraints, affected evidence items, changed margins, and whether the current plan remains valid.

Assessment domains include: energy balance, generator feasibility, BESS SOC and power capability, operating reserve, forecast uncertainty, bus voltage, line loading, transformer loading, and approval status.

6. Plan Validity

MON-010 — A plan shall remain Valid only when material inputs remain within configured thresholds and all applicable engineering constraints remain supported by current evidence.

MON-011 — A plan with an unresolved material change shall be marked Review Required or Replan Required rather than silently continuing under the old assumptions.

7. Monitoring Examples

Event	Baseline	New State	Result
Solar forecast	120 MW	98 MW	18.3% change → impact assessment
Load	180 MW	192 MW	6.7% change → monitor; no mandatory trigger
G1 availability	Available	Unavailable	One unit change → monitor and assess if reserve margin is affected
BESS capability	50 MW	35 MW	15 MW change → impact assessment
Topology	N5 connected	N5 isolated	Topology change → mandatory impact assessment

8. Replan Requirement and Human Review

MON-012 — A replan shall be required when a material change causes a generator, BESS, reserve, voltage, thermal, or forecast constraint to become invalid or unsupported.

MON-013 — Human review shall be required before executing a materially changed plan that has not been revalidated.

MON-014 — Threshold values in this document shall be configurable parameters and shall not be treated as permanent engineering policy.

9. Tool 11 / Tool 12 Demonstration Mapping

Tool 11 concept: monitor the current operating snapshot and identify whether a plan-change trigger has occurred. Tool 12 concept: assess the impact of the change and determine whether revalidation or replanning is required.

The tools are demonstration concepts. This document provides the synthetic engineering evidence and workflow vocabulary needed to test their integration.